Computer Science

Csc 343

Final Project

Due by December 12

Instructors: Donald Lushi, Fady Soliman

In this project we will use all the components that we have built so far to design a circuit capable of adding 10 numbers loaded from RAM.

TASKS TO DO:

1. Create a 32 words SRAM with word size 32 bit. Initialize it with 10 values using a mif file.
2. Create a 32-bit add_sub circuit.
3. Create 32-bit register.
4. Create an Address Control Unit, that holds the initial address 0 and increments it every time increment signal is 1. The reset signal resets the address to 0.
5. Using components like Flip-Flop, MUX, DEMUX, Address Controller connect the RAM, registers and add_sub together as shown in Figure 1. This circuit can perform addition/subtraction with any number of values.
6. Write a testbench that tests the circuit by reading values from RAM and:
    1. Adding 2 numbers
    2. Subtracting 2 numbers
    3. Adding 10 numbers
    4. Subtracting 10 numbers
    5. Store the results of each of the previous operations to RAM.
7. RAM, registers and Flip-Flops should all be synchronized by connecting them to the same Clock.
8. Run the simulation to generate the waveform. The simulation should show input signals and correct results.
9. Change one of the values in RAM to a big enough number that causes overflow during addition.
10. Run the simulation again to check the overflow.
11. Introduce an error in your circuit and run the simulation again. Your testbench should detect the error.
12. Document your work and submit the report and a zip file with your code. You will have to describe your work on December 13^{th} during class. Make sure your code is also ready to run in case you are asked to run it.

Note: Signals Reg1Load, Reg2Load, Reg3Load, Reg4Load, WriteEnable, SelectDemux, SelectMux, AddSub all have a 1-bit size.

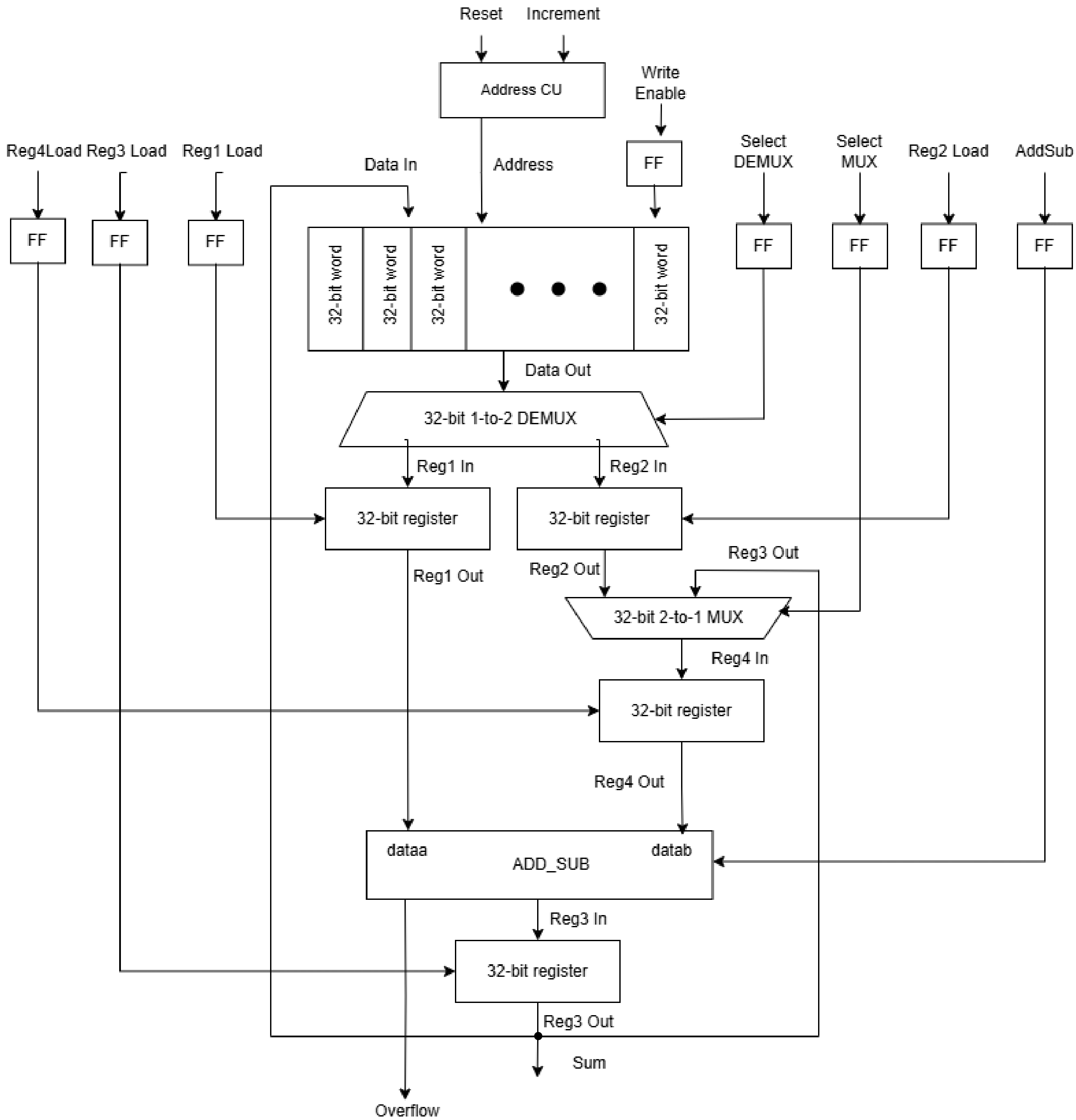

Figure 1. Circuit diagram